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# Industrial Vision Systems

*COGNEX In-Sight Explorer®*

MCR²

Manchester **Robotics**

# Industrial Lighting Vision Systems

## COGNEX®

- Cognex Corporation® is an American manufacturer of machine vision systems, software and sensors used in automated manufacturing to inspect and identify parts, detect defects, verify product assembly, and guide assembly robots.
- Global leader in machine vision systems
- Used in:
  - Manufacturing
  - Quality inspection
  - Robotics and automation
  - Focus on reliability, accuracy, and speed

Tutorials and Training:
https://www.youtube.com/watch?v=kg3nAY024bA&list=PLaDbiumRhU8SSFlB5s0m4jlsNCFN7xUQW

# Introduction

## In-Sight Explorer

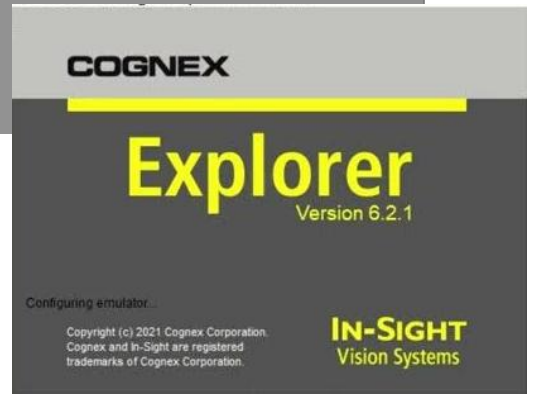
- A **PC-based software** for configuring Cognex In-Sight cameras
- Used to:
  - Set up vision inspections
  - Program image processing tools
  - Monitor camera results in real time

[Tutorials and Training](): https://www.youtube.com/watch?v=kg3nAY024bA&list=PLaDbiumRhU8SSFlB5s0m4jlsNCFN7xUQW

# Introduction

## In-Sight Explorer

- Capture and display live images
- Detect features such as:
  - Edges
  - Patterns
  - Text (OCR)
  - Presence/absence
  - Measure distances, angles, and positions
  - Make pass / fail decisions, etc.

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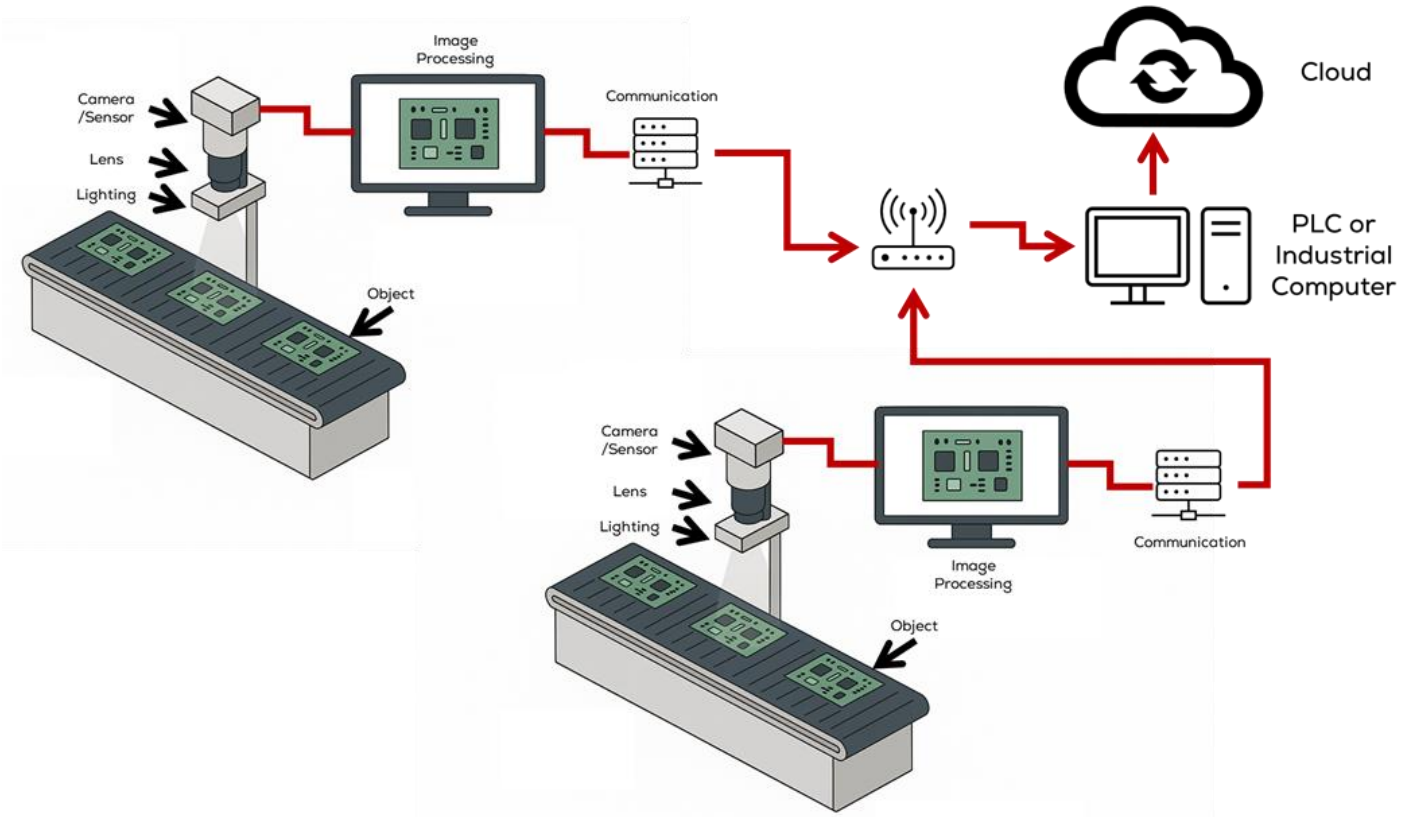
# Introduction

## Example Basic Workflow

- Camera captures an image
- Vision tools analyse the image
- Results are evaluated (OK / Not OK)
- Output is sent to:
  - PLC
  - Robot
  - Production system

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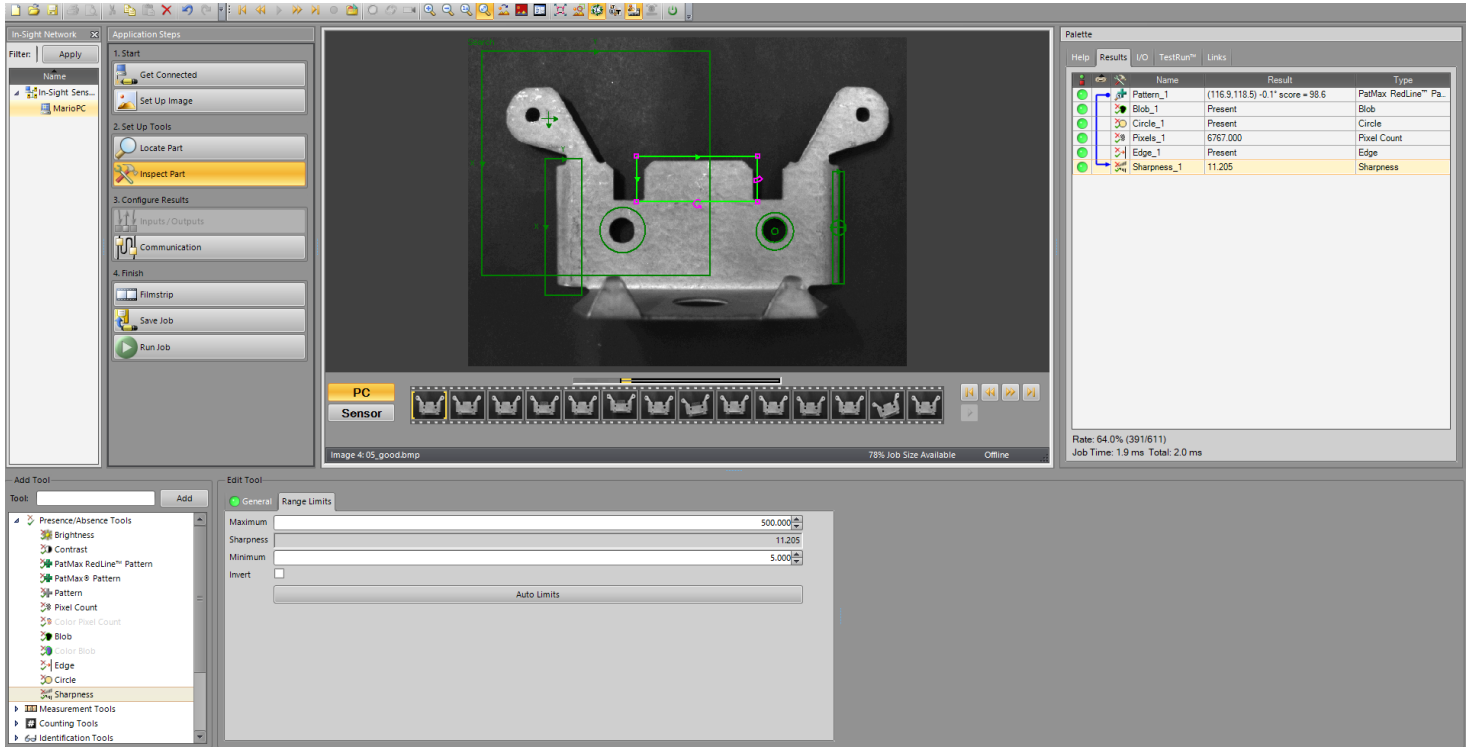
# Introduction

## Programming Approaches

- Two Ways to Work in In-Sight Explorer
- Spreadsheet View
  - Visual, logic-based configuration
  - Similar to Excel
  - Very popular in industry
- EasyBuilder Interface
  - Step-by-step guided setup
  - Ideal for beginners, technicians, etc.




[Tutorials and Training](): https://www.youtube.com/watch?v=kg3nAY024bA&list=PLaDbiumRhU8SSFlB5s0m4jlsNCFN7xUQW

# COGNEX In-Sight Explorer®

*Installation*

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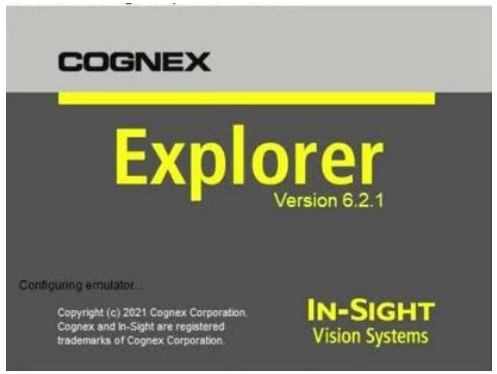
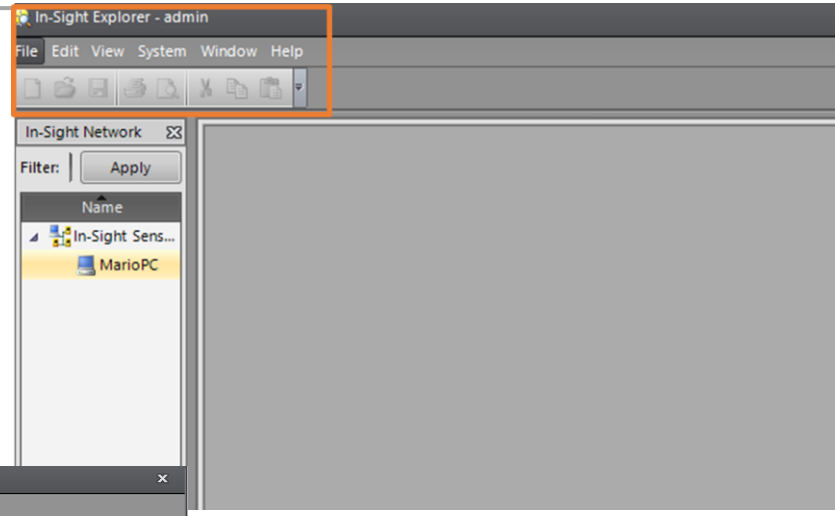
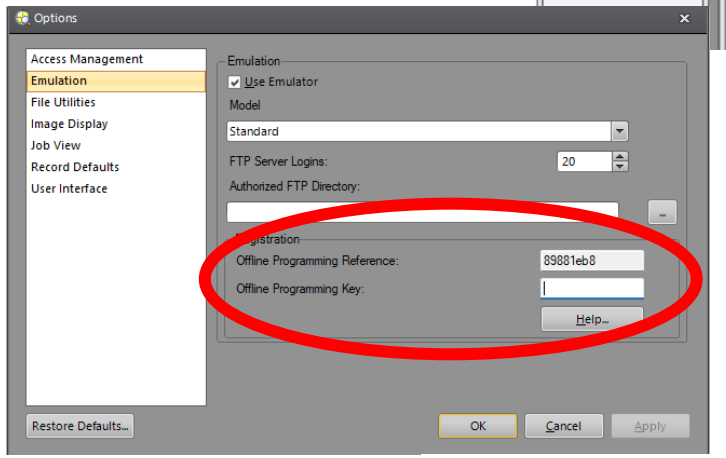
# Installation

## Installation (Emulator)

- Download the In-Sight Explorer 6.5.0 from here
- Unzip the folder and run the Setup “Cognex In-Sight Software 6.5.1”
- Once the installation is finished run the software
- In the top menu go to *System>>Options*
- Click on Emulation Tab
- Go to the following website and create an account.



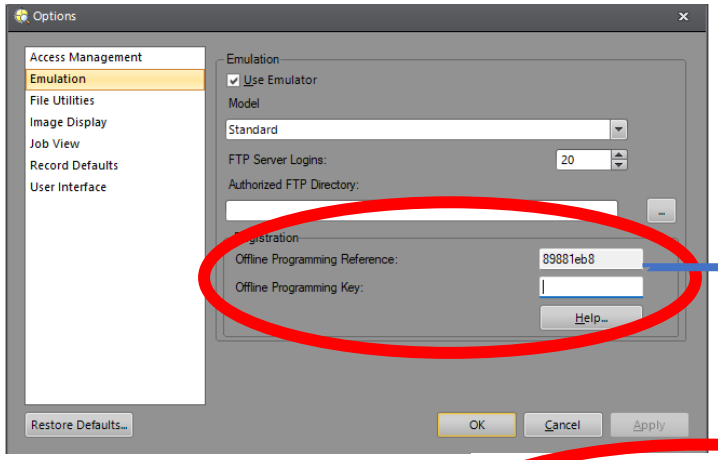



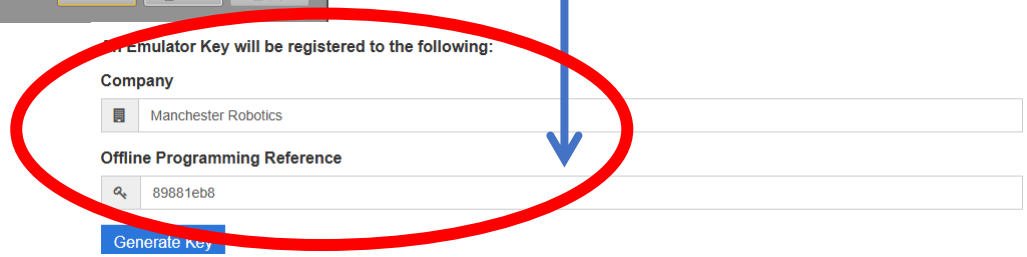



# Installation

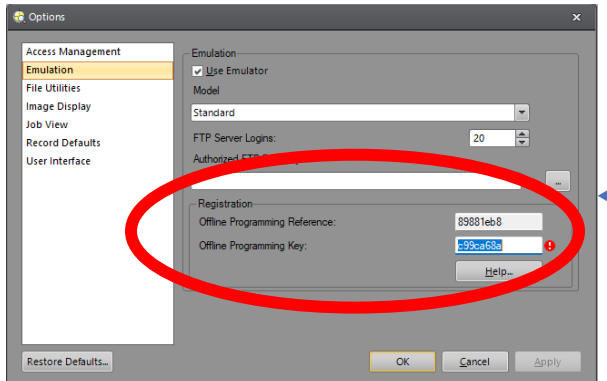
## Installation (Emulator)

- Go to the bottom of the page and <u>type the name of the school</u> as the *"Company"* and paste the *"Offline Programming Key"* from the options menu.
- Click *"Generate Key"*
- Copy and paste your generated key into the "Offline Programming Key" in the options menu.
- Click "Apply" and "OK"
- Close the program and open it again



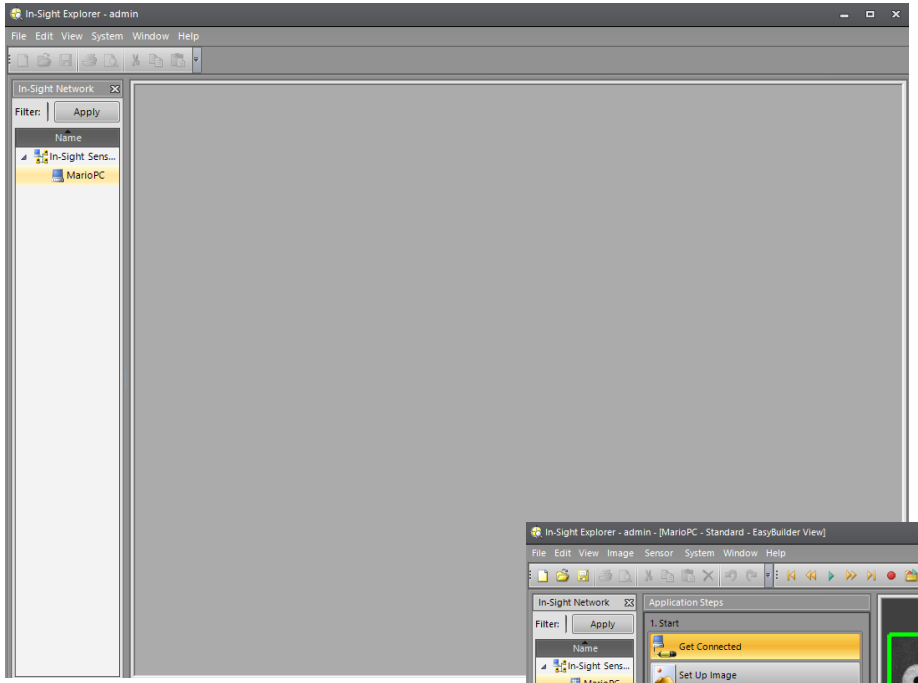

Your In-Sight Emulator Key is **c99ca68a**. Please save for your records.

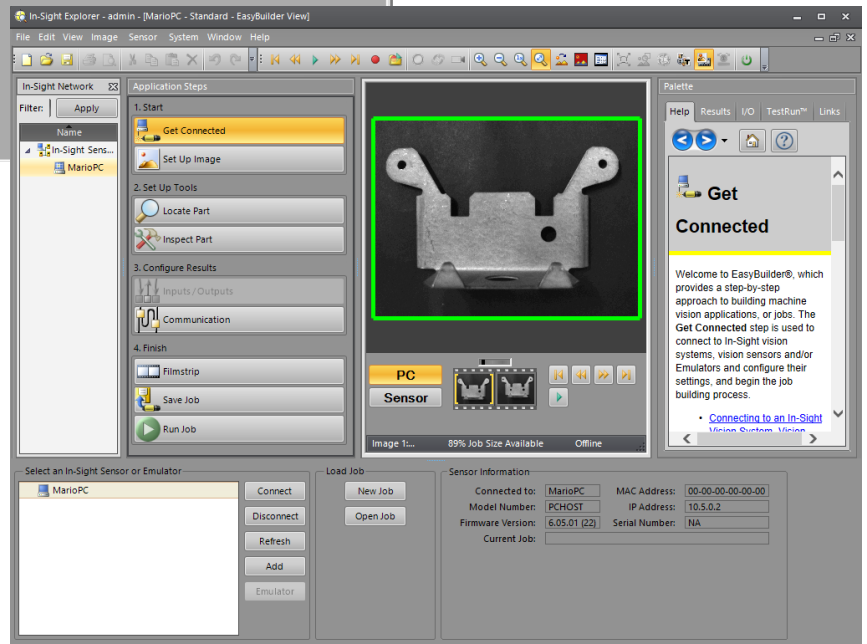


# Installation

## Installation (Emulator)

- The program should look like the image
  - If the workspace starts greyed out, click View>>Insight Network
- Double Click on the name of your computer
- Your scene should look like figure 2 (without the central image)
- Cognex Insight has been successfully installed and configured as emulator.



# Thank you

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# T&C

*Terms and conditions*

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